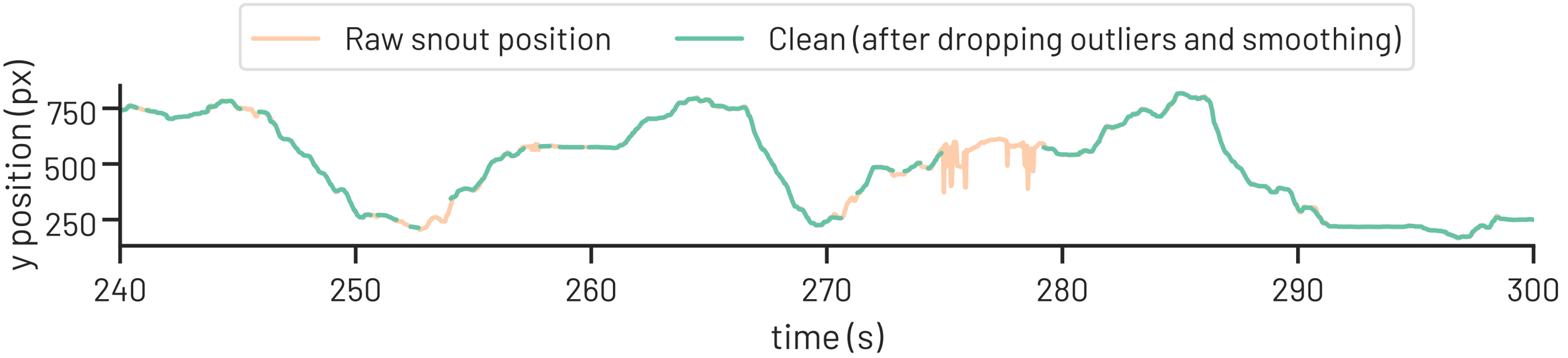
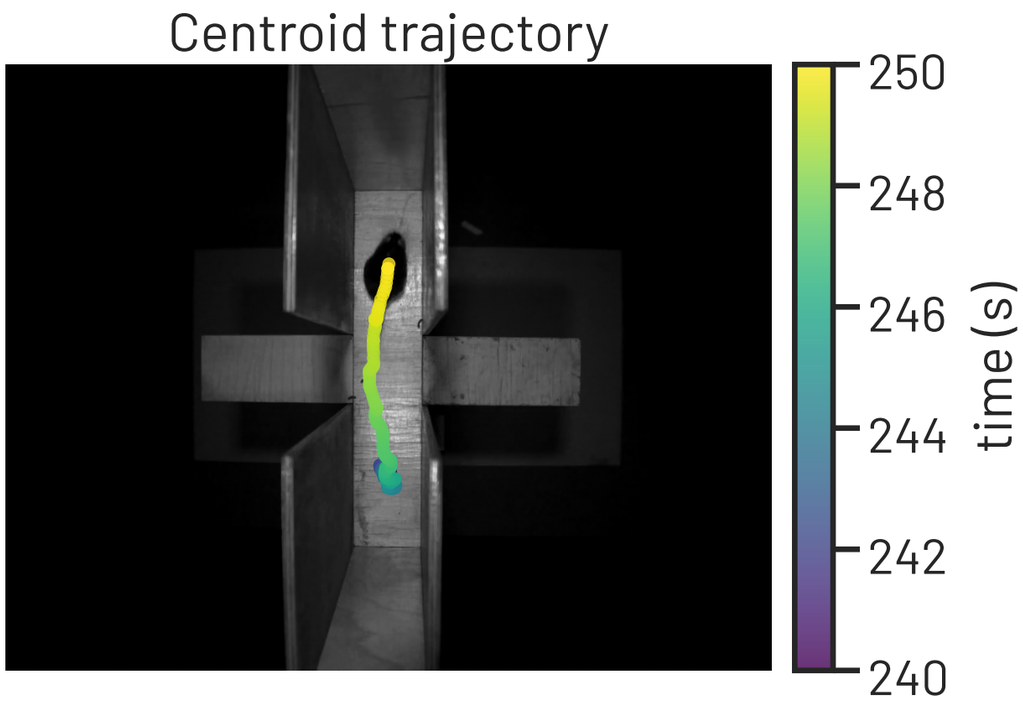
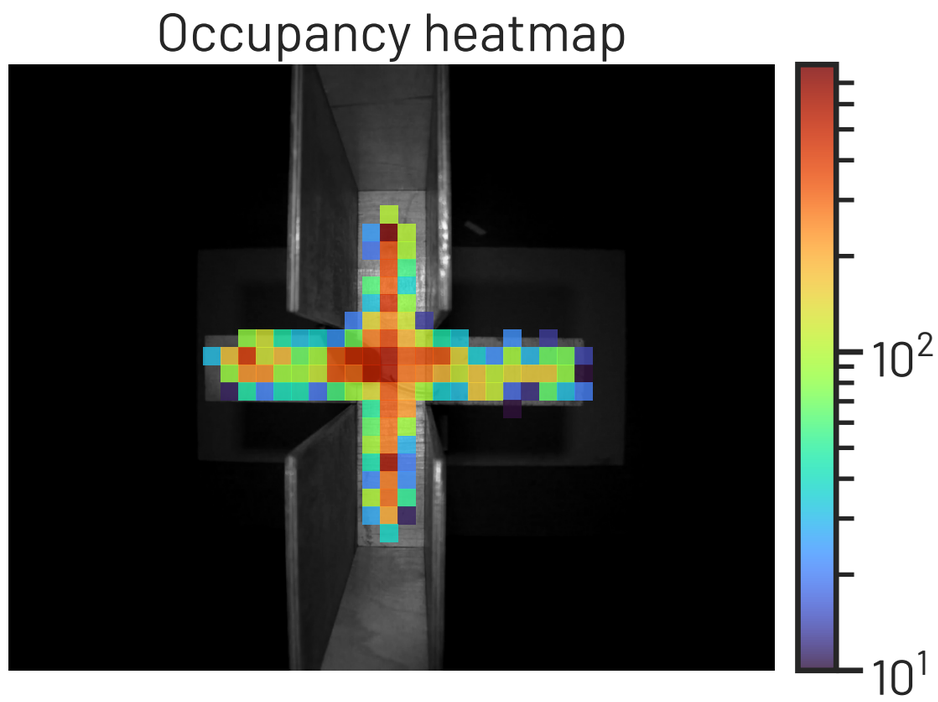
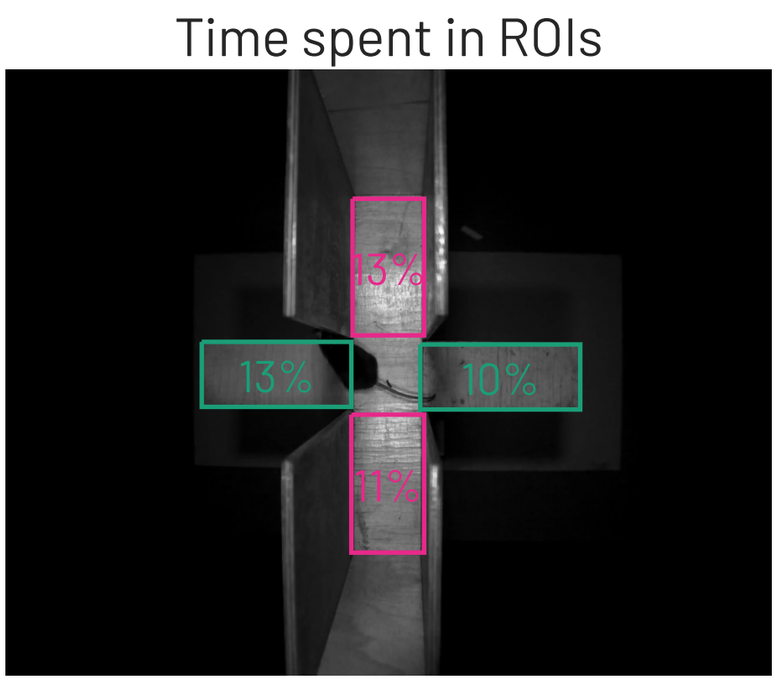
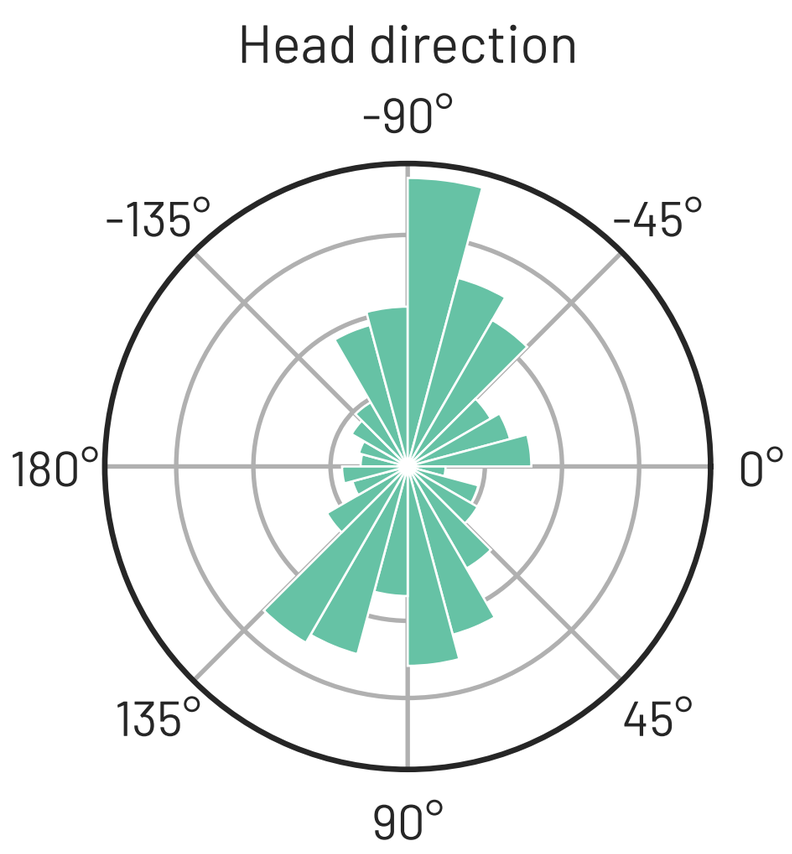
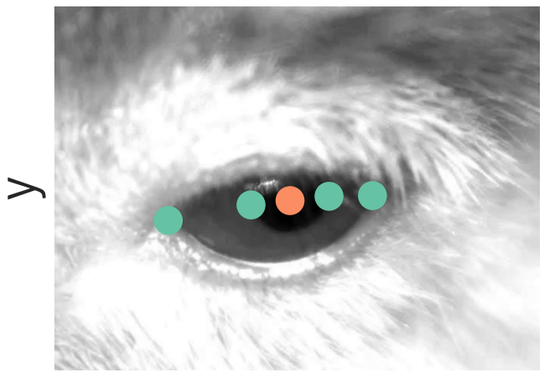

Raw snout position
Clean (after dropping outliers and smoothing)
y position (px)
750
500
250
240
250
260
270
280
290
300
time (s)

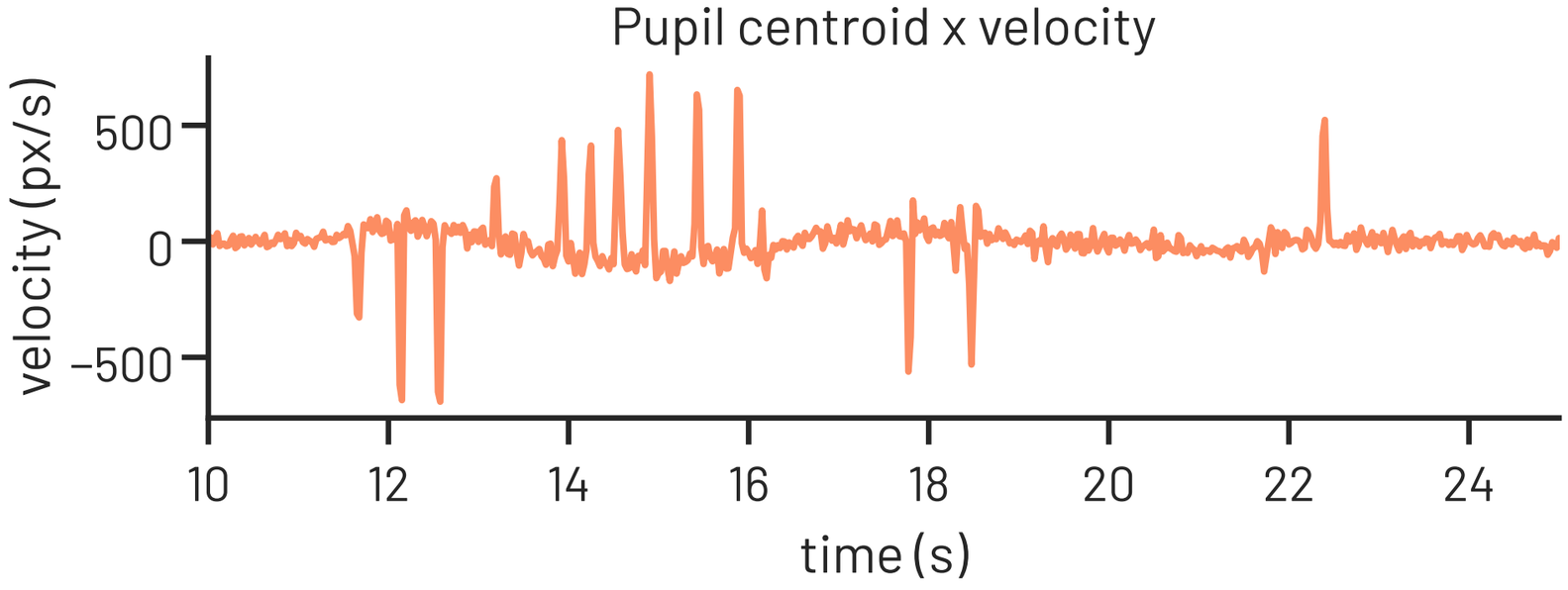

Centroid trajectory
250
248
246
244
242
240
time (s)


Occupancy heatmap
10^2
10^1
frames


Time spent in ROIs
13%
13%
10%
11%


Head direction
-90°
-135°
-45°
180°
0°
135°
45°
90°


y
x


Pupil centroid x velocity
velocity (px/s)
500
0
−500
10
12
14
16
18
20
22
24
time (s)